

Emmy Blumenthal — Curriculum Vitae

Pronouns: They/them/theirs

Princeton, NJ || Salt Lake City, UT

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 0009-0002-1409-9550

Education

Princeton University

Sep. 2024 – Present

PhD in Physics (affiliate of the Center for the Physics of Biological Function)

- Advisor: Dr. Gautam Reddy

Boston University (BU)

Sep. 2019 – May 2023

B.A. in Physics (with Honors) and Mathematics

- Thesis: Typicality and Chaos in Asymmetric Consumer-Resource Models (Advisor: Dr. Pankaj Mehta)
- GPA: 3.91, Magna cum laude in College of Arts and Sciences

Honors and Awards

- [Hertz Fellow](#) (2024–2029)
- [National Science Foundation Graduate Research Fellowship Program](#) (2024–2027)
- [Princeton Centennial Fellowship](#) (2024–2029)
- Sigma Xi Member (2025–)
- [American Physical Society Division of Biological Physics Travel Award](#) (March Meeting 2023)
- [BU Department of Physics College Prize for Excellence](#) (awarded to one senior in each department)
- [BU Trustee Scholarship](#) (full-tuition, merit-based)
- [BU Undergraduate Research Opportunities Project](#) (4 semesters)
- University of Utah Department of Chemistry High School Research Award

Scientific Papers

- Zhijie Feng, [Emmy Blumenthal](#), Pankaj Mehta, Akshit Goyal, *A theory of ecological invasions and its implications for eco-evolutionary dynamics*, [PNAS](#) 122 (49) e2505850122 (2025).
- [Emmy Blumenthal](#), Jason W. Rocks, Pankaj Mehta, *Phase transition to chaos in complex ecosystems with non-reciprocal species-resource interactions*, [Physical Review Letters](#) 132, 12740, **Editors' Suggestion** (2024).
- [Emmy Blumenthal](#), Pankaj Mehta, *Geometry of ecological coexistence and niche differentiation*, [Physical Review E](#) 08, 044409 (2023).
- Deniz Aybas, Janos Adam, [Emmy Blumenthal](#), Alexander O Sushkov, et. al., *Search for axionlike dark matter using solid-state nuclear magnetic resonance*, [Physical Review Letters](#) 126, 141802 (2021).
- Matthew J. Kummer, Yoo Seok Lee, Mengwei Yuan, Bassam Alkotaini, John Zhao, [Emmy Blumenthal](#), Shelley Minter, *Substrate channeling by a rationally designed fusion protein in a biocatalytic cascade*, [JACS Au](#) (2021).
- Min Li, Kevin Klunder, [Emmy Blumenthal](#), Matthew B Prater, Jack Lee, John E Matthiesen, Shelley D Minter, *Ionic liquid stabilized 2,2,6,6-Tetramethylpiperidine 1-Oxyl catalysis for alcohol oxidation*, [ACS Sustainable Chemistry & Engineering](#) (2020).

Research Presentations

- [American Physical Society March Meeting](#), *Minneapolis, MN* (2024)
“Phase transition to chaos in complex ecosystems with non-reciprocal species-resource interactions”
- [American Physical Society March Meeting](#), *Las Vegas, NV* (2023)
“A new geometric framework for niche theory and consumer-resource models”
- MIT Physics of Living Systems Short Talks, Invited Speaker, *Cambridge, MA* (2023)
“Typicality and dynamical fluctuations in nonreciprocal consumer-resource models”

Research Experience

Biophysics Theory Group (Dr. [Gautam Reddy](#)) || Princeton **Oct. 2024 – present**
Graduate student researcher

Biophysics Theory Lab (Dr. [Pankaj Mehta](#)) || BU **Dec. 2021 – Aug. 2024**
Undergraduate Researcher (UROP funded) until May 2023
Full-time Post-baccalaureate Researcher from May 2023 to Aug. 2024

Quantum Lab (Dr. [Alexander O Sushkov](#)) || BU **May 2020 – Sep. 2021**
Undergraduate Researcher (UROP funded)

Bioelectrochemistry Lab (Dr. Shelley Minteer) || University of Utah **May 2018 – Sep. 2019**
High School Research (Award funded)

Systematic Randomness Testing || [Wolfram Summer School](#) **Jul. 2019**
Research Project

Teaching, Mentoring, and Outreach Experience

Physics Department || Princeton **Spring 2026**
EDI Fellow

Physics Department || BU **Sep. 2022 – May 2023**
[Learning Assistant](#) — CAS PY 451, 452 (Quantum Mechanics I & II)

Faculty of Computing and Data Sciences (CDS) || BU **Sep. 2022 – May 2023**
Lead tutor — CDS DS 122 (Foundations of Data Science III)
Tutor — CDS DS 121, 210 (Foundations of Data Science II, Programming for Data Science)

Wizards After School Program || BU Community Service Center **Sep. 2022 – May 2023**
Volunteer

PeeRs for Incoming Student Mentorship ([PRISM](#)) || BU **Sep. 2021 – May 2022**
Mentor

Chapter of Society of Physics Students ([Photon](#)) || BU **Sep. 2019 – May 2023**
Member, occasional presenter